

Manufactured uncertainty: protecting public health in the age of contested science and product defense.

.

The strategy of "manufacturing uncertainty" has been used with great success by polluters and manufacturers of dangerous products to oppose public health and environmental regulation. This strategy entails questioning the validity of scientific evidence on which the regulation is based. While this approach is most identified with the tobacco industry, it has been used by producers of asbestos, benzene, beryllium, chromium, diesel exhaust, lead, plastics, and other hazardous products to avoid environmental and occupational health regulation. It is also central to the debate on global warming. The approach is now so common that it is unusual for the science not to be challenged by an industry facing regulation. Manufacturing uncertainty has become a business in itself; numerous technical consulting firms provide a service often called "product defense" or "litigation support." As these names imply, the usual objective of these activities is not to generate knowledge to protect public health but to protect a corporation whose products are alleged to have toxic properties. Evidence in the scientific literature of the funding effect--the close correlation between the results of a study desired by a study's funder and the reported results of that study--suggests that the financial interest of a study's sponsors should be taken into account when considering the study's findings. Similarly, the interpretation of data by scientists with financial conflicts should be seen in this light. Manufacturing uncertainty is antithetical to the public health principle that decisions be made using the best evidence currently available.